ActInf Textbook Group ~ Cohort 6 ~ Session 21 (Applying Active Inference & Ontology) 9/9/2024

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all right September 9th 24 and Andrew go for it um hi everyone so uh as we just briefly I know some is coming in uh as we just briefly discussed uh since today is another session where we are not directly talking about any particular the chap chapter of the textbook but rather um just talking about applying active inference more generally um we agreed that I would spend just a little bit of time here uh giving sort of a quick rundown of how to construct an active inference agent in this case it'll be specifically uh a PDP agent meaning we're working with um well in this case we're working with discrete time and discret observations um as opposed to continuous time models which you'll see in chapter 8 of the textbook um and I kind of just want to focus on just from scratch like how do we think about constructing an agent um I won't be touching on how to construct an environment um but but that's the kind of thing that you know you would you would bear in mind whenever kind of coming up with an idea for a simulation or experiment you want to run um and actually on that note Daniel is it possible for me to um share my screen by chance yep go for it awesome thank you so this was the um and if anyone wants this link they can find me on GitHub a p hea just like my last name on the zoom uh and the repositories called ic2 S2 active inference tutorial um this is material that I presented back in July at University of Pennsylvania um I basically uh gave a tutorial on AC inference how it might be applied to social sciences specifically in uh constructing multi-agent simulations so um and is this uh is this visible to everyone the slides here yeah go for it for you know five to 10 minutes however much you just want to footnote and then let's see what people want to continue with definitely sounds good um so in this tutorial along the way I you know we're introduced active inference there's some nice um explanations of different concepts from the textbook Etc talking about uh the basy brain hypothesis General ideas how Bas rule kind of plays out in active inference and eventually we reach where uh we're going to use the pmdp package um which has been developed to make discrete time AC inference agents in Python and um there are a lot of updates being made to that package right now if you were to like use a git clone to download it directly there's a lot going on in there but for the time being I'm going to just be using the the PIP install um Pi Pi version uh it's a little bit more kind of uh packaged up comes as more like a single unit so um whenever we construct active inference agents uh the starting step should be in my view starting step should be what are the hidden states that the agent has in their generative model like what are they trying to infer uh second thing is what are the observations that the agents will encounter in the world and then three what are the actions uh or controls or policies those three terms are sometimes used interchangeably they don't necessarily mean exactly the same thing um but for now we'll just go through with all of those we'll call them actions uh what actions can the agent commit uh so so we can view that kind of like like this like the this is the agent that we be constructing um all of these things play out it's called an action perception Loop um let me see if I can just okay hopefully that's visible enough to everyone um so so that we'll we'll start at the bottom of this action perception mod um first observations there are two kinds of observations that this agent will receive at any given time step um quick note that this entire action perception Loop plays out at every time step we're working in discrete units of time and so you know whether it's from second to Second minute to minute millisecond a millisecond however um all of the units of time are the same uh here the agent receives one of two observations one of them is related to uh an improve Improvement if something improves or if it doesn't improve and I didn't include a third category but there's another one called unobserved just bear with me and we'll go into what that means soon um but that's one particular kind of observation otherwise known as an observation modality that the agent has slashes we can think about about this is like in the same way that we have you know sensory data that comes in uh you know like you have cold receptors on your on your uh in your scan on your arms so you can feel if it's cold or it's hot or otherwise similarly here the agent has some hypothetical receptor and we could call it visual or otherwise but the point is they they get one of these three observations Improvement no improvement or an observed um another observation modality they have is self or neighbor and I call this the attention modality uh at any given time step they're paying attention either to themselves or to their neighbor um part of why we're talking about self neighbor that kind of thing is that eventually this leads to a multi-agent simulation where where agents can can either uh kind of do their own thing or they can interact with other agents and so this attention modality means they they're actually observing like who am I paying attention to um so when they take in those observations they have a single hidden State uh which is I've denoted s with superscript ATT this is their attention hidden State um an agent takes in observations and then using what's called a likelihood model which is uh uh in in in chapter 7 we refer to it as the a matrix remember we have the the a b c and d matrices uh this is the a matrix and so the a matrix is this it's the probability of that observation that they just received given a hidden State uh whenever the agent reaches this this right uh this Square on the right so to speak this stage of action perception cycle uh they invert their models so now they they actually try to figure out what is the hidden State given I just received this observation you know in the same way that you could receive a a hot or cold observation and so you infer oh it's it's cold here um here the agent is for example they might observe themselves and then infer oh I'm paying attention to myself um and vice versa for neighbor here this is an incredibly uh despite the material despite kind of the complexity of active inference Theory this agent is actually very simple they can receive one of two observations every time step they have a single hidden state that they're trying to infer uh self or other but what that does is that it makes a link between if they observe an improvement or no improvement uh with who they're paying attention to and whenever you combine that with the idea of learning their likelihood model which is what this Delta this triangle means like they're learning their likelihood model um they can learn that oh if I pay attention to myself I see an improvement more than if I pay attention to my neighbor uh right so so so these kind of values get these parameters get adjusted over time these are the agent's beliefs changing based on their learning from experience and then after they make an inference they then decide okay which action should I take on this time step uh the names of the actions are explore and exploit which is um that doesn't quite have anything to do with the the so-called explore exploit dilemma that comes up in the textbook um you know that's technically a different thing these are just arbitrary names it explore here really just means the agent is explor exploring new Solutions this is a problemsolving agent the idea is they can explore meaning on their own pay attention to themsel and look try and come up with new solutions to a problem or they can exploit meaning that they will steal the answer of their neighbor right and if they so so you can see how if they steal the answer from their neighbor and they see an improvement then they start to learn like oh if I just keep taking my neighbor's answer to this problem I keep seeing improvements or it might be the other way around I might see no improvement start to associate like ex the action exploit receive the observation neighbor and no improvement and then in their likelihood model they'll learn like oh I shouldn't steal from my neighbor that least to you know know progress on me trying to solve this problem now I've already discussed these three kind of squares these three not of of this graph here um that's all you need for an agent uh the only thing I've I've left out is the environment and it's the environment that you know for a a programmer you can think like oh well if I set up an agent who has observations and in state to infer in actions that just gets hooked up with the environment the environment can be a list of you know if else roles like if the agent does this then the environment should send out this um or it could be more of a stochastic process you could use you know a random number generator or otherwise um so the agent will learn over time so if anyone is uh interested in this again feel free to to check out the uh I'll actually put the link in the chat whenever I'm done here I don't want to take up too much time though but it's just that's the general process of an action perception Loop and then you know the tutorial from there just kind of starts to break it down even further like once you so once you figure out those hidden States like what is the agent trying to infer what are the observations that can receive what are the actions that can commit The Next Step would be okay how do we link those three things together through your a b c d and e um you know a links together observations States be links together transitions between states with what the agent does it's like oh if I think X right now and I do y then I believe that c will happen next that's what kind of like what a b Matrix says C Matrix prior over uh observations which is like your preferences for what you want to see the agent will kind of try to realize these or seek out these observations by way of free energy minimization uh D is your initial beliefs about States you know what did you think before um that and then and then through uh inference we'll we'll end up with a state uh belief a posterior that might differ from D or it might be exactly the same in which case that means that the agent just reconfirmed its own belief again it's exactly the same and E just your priors over actions so what are you most likely to do regardless of anything uh that's why they're also known as habits um I'll I'll stop that there's there's plenty more to go into but those are all the core barebone aspects of what you need for an agent and the rest is just you know I pretty quickly start jumping into code from there on how to do it so um yep that's it for now so oh sorry can I have the link to that I'd love the link to that in absolutely yeah worries I mean that's right my wheelhouse I love this stuff bu and I I just spend all my days doing this kind of things so I'm it's nice to see somebody else like building out like the step-by-step process of like how do we like make this work kind of and then you get to like just throw it all you know that way other people can do it too yeah cool yeah yeah exactly the point yeah I like for me personally it's like it's only uh I'd already been involved with the institute for some time and then realized like there is no clearcut way of do doing this um I just dropped the link in the chat or sorry uh it's not the specific um but in any case yeah so so what I did was that I just started getting the documentation for PDP and I did a lot of like just going through the repository to make sure I'm getting things right um there's a lot of complexity because they tried to build imdp off of SP m in mat lab which is what Frist and other folks developed and it's just been in development for years so they're trying to recreate this massive project and uh I just think like well if you want to get more people using it let's make some more kind of from scratch uh tutorials on how to go about this so that's that's what I did but cool I'm glad glad you find it interesting oh yeah I mean I had to come at it from a complete different uh uh hold on I get back in this thing here I lost all my stuff um yeah the uh I came at it from a completely different angle and but came to the same conclusions and it's I had no idea that somebody was doing that it's so much fun there we go there's everybody yeah there there's a lot to add I think one of the biggest pieces then anyone who wants to ask go for it is like what is the question about an active inference generative model what's a question that staples into A B C or D or a line of code or a specific node or Edge on a graph what's a question about the world so it's like what's a question about the subway map of the city what is a question about the territory of the city that can get back and forth and we know all about and talk all about these kinds of ambiguities and and dual reference and things like that but at the same time it is a different type of question to say what is the dimensionality of the a matrix in comparison with what is the dimensionality of aent sense making or something like that and to know when one is clearly on one side of the other or when you're not sure is some of the fastest ways to start separating kinds of questions according to is this something that's just going to be addressed in a line of code or a math equation or is this something that's more like a real world line of inquiry that cannot be answered in any string of words that somebody says I feel like if you can't answer it then it you know it's like that I mean there's a lot to be said for not being able to answer something right like um but uh I definitely have a lot of questions um I'm excited to read the material so I can like start asking those in a little a little better way I'm trying to think right now yeah Zach go for it oh I was just gonna ask if you guys think are there any questions that you fundamentally can't answer or questions that we just don't have the ability to answer this might be a little bit more philosophical but I feel like Daniel you were kind of leading to potentially the question of what is it like to be a bat and maybe ideas that kind of into that realm talking more about something about the agent and its relationship to the environment instead of not instead of anything I'm sorry that's the end of the sentence I just want to say really quickly that uh this reminds me of um a really crucial point they make in the textbook uh I I think it's in chapter nine but there's this uh early on in the chapter it it must be chapter nine but I might be mistaking that but it has to do with analysis and the whole point is they have this figure that shows like okay on the innermost uh kind of square of the figure we see the typical like graphical model that they you know A variation of that they keep showing through the textbook but then on the outside like there's a square around it denoting like okay it's really making the point that this is what we the scientist or experimentor think is the model of the agent yeah thanks Daniel that we're trying to develop right it's just like really trying to make the point that like we ourselves don't firmly know necessarily what the inside of the mo like a rat in a teammates like I'm trying to model it I have to be aware that like here's the level that is what I think is the rat's perception of what's going on and how it learns and so on but you know so it's like the objective model is just like outer like this is what I'm constructing like you're in the picture too is how I yeah great point and and for examp ex Le the experimental actions of the experimentor are just it's just taken as an observation train we're not modeling the policy that led to the experiment being designed okay but you could include that so you can continue extrapolating um but it it also makes clear where where the the buck stops but about the questions it can't be answered I mean big question right but there's a sense in which behaviorally every question is always answered if with acknowledgement a postulational head not dare I say or silence um or or or some other kind of like nons respon so you know can be answered in Trivial sense yeah everything already is can be answered in more like a Justified true belief semantic validity sense so here's a few pieces from the ontology project built up over the years but I mean just a snapshot if someone wants to dive in further and add more it's the greatest types of contributions that can help at this on logical level here's like many many many of the phrases and writings in the textbook group are like how is this term related to this um when they said this did they mean that does this mean this what does xers think does X why would X person want to believe that this was an explanation of that I mean whether or not you go into it questions have these templated forms so clarifying like what is active inference what surprise um going into those and knowing that even with the awareness of the question like what is surprise or how are they using surprise here those are already categories of questions that you can go in with a structural map towards and another fun um Direction with the with the philosophical side with like Oli and and Paulo Maria and others in the ontology was like starting with the Linguistics of the you know how could we ask and proceed on a question without like knowing what was written from a symbolic perspective so it's like okay then what is the the the lexical semantics like did they say something in the paper here we here we Define a for here we use surprise to mean this or here we operationalize surprise as so it's like they're narrowing down or leaving implicit what they mean by surprise but if they said we by surprise we mean one and then a person constructs an argument where it's predicated on surprise being number two then it's off base then getting into their local weaving of the relational semantics so not just how they Define the terms but it's like okay given that they said blanket states are sense and action you know then did it make sense with this other usage of set that's in the internal reading internal consistency view from inside like was there an apparent issue without even going a view outside of it then getting into views from the outside oh I think that they didn't do this or not I don't think enough attention was paid to this they should have done this I think it was cool that they did this um what would an embodied person think about this like all manner of then externalized but it's kind of just like a combination of like a close and deliberate and charitable structured reading in that with reference to the AC in onology but it could just be any ontology system at that point you give the internal reading and understand when questions and methods are being used that are related to internal reading and then when is it like an external reading from like a different perspective or a later time what do you think Zach or what what kinds of questions does that does that answer or not I mean that makes it seem I really like the external idea of looking at it for many many different lenses because then we're naturally going to find ways to maybe collapse some language and find other things that might have been equal in places that we weren't expecting to look I think just translating into a different language can have a ton an amazing amount of positive impact and it just makes collaboration a really cool thing but at the same time I I've always been attracted to active inference because it has this uh physics backing it all it all comes from the principle of least action and everything about that so in my head there should always be that physical explanation for all things or at least a mathematically based explanation for all things which really speaks to me personally man if I can jump in here I really appreciate that because it really speaks to me personally too like being able to pull things out of the out of the the discourse and put them into theoretical Frameworks mathematical Frameworks gives us it's like that low and high road that active inference you know it gives us a path to creating tangible Solutions around these things and I I think it's like it's kind of esoteric almost in a way I mean the you know Daniel uh I think and I were talking last week about how for instance it's this is so valuable for people who love all manner of Sciences because they're like uh it it intersects everything philosophy and on and on and on you guys know um and um it does I feel like it necessarily requires like atic perspective an external perspective to um and objective perspective to like write about these things meaningfully you have to touch base on everything and you guys should see what a lot of this stuff does when you uh put it into different uh indigenous languages pretty yeah you're absolutely right the formalism 100% there so many ways to go with but like modal logic like the shirt might be blue or or conditional Logics or paraconsistent all these other kinds of of alternative logical structures where things are not like either one way or another outside of an evaluation Etc it's like if someone says well ramstead claimed X and I think he's saying it's like that cannot be an incorrect second half of the sentence unless it were were deceptive but if it were deceptive like the person was like I think that ramed means one but I'm going to verbalize that I think he meant to that still would be their veritical behavioral representation so it would truly be what they had said at that time and they can have some other side information like um but I said that because of this or I would have said that so it's like but they can't be wrong so it when the space of ideas is so vast there's so many PES more than we'll ever read Plus+ plus generative intelligence generating the dissertations we don't have yet on demands and and more expressions are not opposed to each other even when they contain what appears to be like the clearest possible like this person thinks that the blanket means X this person says it's not X and their their epistemic frames are all aligned it's like even in that kind of idealized scenario all the quantum reference frames were aligned we had gotten down to it and then two people differed and one said it was up and one said down still that situation could be analyzed as a cognitive environment where they both truly had those perspectives and then a third Observer could take another take on it and it would also only add to the situation to piggy back off of that I think that's exactly what we saw in geometry in the early 1900s with the airong program where essentially there were a bunch of geometries there was a parabolic geometry which I don't think that's how you pronounce that now that I've said it uh there's uh this algebraic geometry all of these different geometries that all seem to talk about very different things they all went in and were able to describe certain theorems but they all took all of these amazing paths around and none of them seemed to intersect until the ER longan program which was able which was of course the mathematician erir longan or was it the place airong in forgive me it's been a minute since I've read it it was the birth of gauge Theory they were able to take all of these seemingly disperate ideas and find a relational way to talk about all of them which gave us gauge Theory today which actually we're seeing in a lot of places it led to Something in machine learning as well called the 5gs which I won't get into detail but it is the main method that we have for trying to understand learning itself as a method to compress as a method of equating intelligence with compression into a different space which is just amazing it really does remind me of kind of the method of scientific revolutions if you guys are into Dusty old books from the 60s I love Dusty old books man ER Paul the gauge Theory like also if we think about the recent development of actin so it's like 2022 textbook it's like oh yeah wait a minute we're in the textbook group 2022 textbook it's like step by step by Smith at all also leveled up with some more context and it's published as a book it's a state-based approach it summarizes and crystallizes the state-based approach to active inference that's why the discussion is like discret State spaces continuous State spaces for variables and for time itself as a variable in the model then especially from 2019 not so much reflected only alluded to in the 2022 textbook coming on much more in papers and in sanj nam josi's future volume two book but the basy and mechanics live stream 49 all of those kinds of things ju opposing the state-based formalisms for active inference textbook discreet continuous Etc with path-based formalisms then ramstead and Dalton pointed towards what they call G Theory and highlighted like different deep dualities and some of their implications as well as the gauge theoretic aspects of what was being discussed like overall so it's a great tie in with with Gage Theory there's so much to to to um build and and clarify around that um so it's interesting that you brought it in but yeah really interesting I'm sorry but uh I think is that the second time you brought up ramstead could you please say a little about him I'm simply not familiar I'm just like I'm just like using him as as like um you know he's he's an active discourse ant and and a great contributor from the pre 20109 PhD student philosophical and his his deep knowledge there and then on through more the applied um and and ongoing um but using it like as sort of just he's out there making or you know friston says X it's just anyone's reported speech um yeah Federated inference yeah yeah that's yeah yeah yeah I was just digging into that oh yeah this was uh I was just sharing this because I'm personally still new to the notion of um paths here and uh I think I think in this paper they replace preferences like like like agents prefer a particular path and to me that that seems to imply like they're not just preferring like particular observation a particular moment in time they're preferring like entire sequence um not sure if I'm mistaken there or not but I was really interested in in that not done reading it yet these are great questions also natural language terms do get overloaded you know with with um with Grievous unnecessary and sometimes requisite overloading and underloading and ambiguity so like a path but then we might say well but the path that they take through space over time one two three we're going to call that a path so then it's like but that's not this one and then you look at from again from an ontological analysis okay paths may or may not depend on action it'd be like wait a minute what how could the path taken in the world not depend on action because path is being used in multiple subname spaces so it be like having a python script where like a variable is defined within an Loop and so it's used the way it's used within a loop but then there's a separate Global context or there there's two functions with the same name so disambiguating the Nam space is some of the most important work each of these can have like a unique um you know star code ID fractal hash so that the local semantics can be described and and and modulated or falsified it's like here's what what we here's where we came down on the semantic Vector for how path was used here does your vector on that usage of path differ and then can can explore like many many more layers of of semantic similarity and difference but yeah I don't I don't know if this is actually using it um in the path integral based basian mechanical sense because this is still a state space based model yeah see yeah it's I I was surprised they didn't go with the word State you know yeah it's what was interesting is uh when you did the the control F and it highlighted all of them the last half highlight there and paragraph the full sentences the remaining tensors encode prior beliefs about paths C and the initial States D and I think that brings up another interesting point of trying to kind of parse through the terms uh in that oftentimes like sometimes the sort of hint of how someone might have been to term is that they related to something else that might be known from an you know whether it be from an entirely different context or not like here they're using the same like alphabetical terms uh to describe like for a it's still a likelihood Matrix for B it's still a transition Matrix but then here C which is typically in in the textbook in chapter 7 in that particular context C refers to your prior over observations but here they're now prior beliefs about paths and meanwhile once again initi prior about initial States D Remains the Same just does a b do so that's sort of like you know it's like this took me off guard when I was reading this I'm like everything here makes sense to me just intuitively right off the bat I've written a tutorial like putting together certain kinds of BPS so uh but then swap out uh preferences or observations for the word paths and I'm not really really sure but um right and then it's like wait but is second and a and City second you know it it's and it's reusing common mathematical notation so it's like the one next level is tremendously informative variable names so calling things with very informative specific names however that becomes unmanageable because you start having long variable names that that just visually make it more confusing and even terms like observation it's like well if it's an action from this then how is it is it are we going to call that an action and the output observation it's like you get into which perspective to take on different sides of Marco blanket for example so one solution that we've worked on in different applications is separating the initialization serialization operations on the variables from secondary assertions about which ontological terms they to and then just making that an explicit data structure and then you can call the variables like specific programmatically generated names and then make programmatic or manual assertions about which terms in a given language you might want to use but calling a term oh we'll do a really informative English name it's like well then you know exactly what namespace that would be informative for whom so if that is the the time cone that someone's working in like within a local for Loop within a programming language within a language Community then the tool gets the job done but that may and then another important thing with with also these uh kinds of sentences is like it is not true without going this whole true thing it's like it it isn't true accurate complete you know all of those things per se you can have a prior that a certain journal or certain co-author certain kind of statement is going to have like this or that kind of probability or attention that should be paid to in all this but it's like best best best case scenario there's no false negatives or false positives and so it's like truth the whole truth nothing but the truth however in in a more probabilistic setting it's like this is just sampled English from a semantic latent object and certainly for for your contri uh contribution to that object or or just grasping it any given finite amount of text is g to have ananus nonacs super relevant you know different relevant realization positives and false negatives Etc with you know so it's like how to both let it just fly by lightly not tripping over every possible fractal epistemic dispute you could raise but and also paying enough attention to enter into their semantic fractal to to see if there is transfer of what yeah and how those Concepts align and yeah I I do you guys feel like there's um uh you know it's like an issue of inference andc conveyance right like um there's like uh all these there's so many brilliant people working on these things and it's almost created like a uh like a you were saying like that disperate kind of ideology it's like we see this in psychology as well especially in indigenous psychology and like diagnosing things like U uh related to like Colonial mindsets and stuff right so it's like there's just so much out there and there's so many things that almost mean the same thing but don't and then there's things that change over time and it's like how do you create like how do you convey like uh a really strict or stringent like pathway of communicating the knowledge while still without like encapsulating the formalism or something and like limiting the creativity and self-expression that people have in this that causes that this field to move forward through Innovation stuff like that right so like for me it's like when I read those words I it's like I I try to ignore the words almost and like just understand um some of the things conceptually because when I go back to my own work those things may be close but they're not the same for instance like you guys uh uh remember that have you read that newpaper uh scale free active inference pixels the patterns scale free active inference right it's like it's awesome I absolutely love it I read it like 500 times when I was in the hospital it's all I had to read so I just read it over and over and over again um and uh but for me in my work uh it isn't uh it's a awesome visual cortex it's like a great road map for a visual cortex and understanding how to like perceive and parse those uh P that perception data which I need for down the road it's not relevant right this moment but um uh but for them it's like it's a it's a pathway to active inference in in a machine um so it's it's there's no like right or wrong way it's so like but at the same time too it's like does then it just fall to what works best or what works first or I mean it's just it's it's tough to navigate certainly yeah definitely you know this is U and all of that's really well said I I mean just speaking on like the the real like particular specificity of like active inference language and stuff another aspect I think about is like like the practices that we have for example like don't plagiarized in academic work leading to every active inference paper that is written that is written for a broader Community than just a very small subset of people that are already familiar with Act of inference it's like well you're going to have to have an introductory section that talks about Act of inference but you cannot repeat precisely what anyone else has said in the same exact way and so you potentially have this kind of uh uh kind of explosion of potential meanings that come out of like you know just a set of a few papers because maybe someone phrased something just a little out of place to where suddenly it could have you know a dozen meanings for that one word now based on how you read the the formulaic constraints and priors on the quote paper form and like expected length tone content all these things it's it's some it's really as that starts to dematerialize and become more computationally augmented more modular all these kinds of things and and just like the one sentence you can now double click on a sentence and say just write the textbook that that underlies this or zoom in on this so it's just kind of like pointing at a continent scale with a course expression or spending more bits of the transmission more more energy and attention to refine and um how that it can change with the way that the papers are written again with the llm based audience shifting all the translational work that we're exploring generating just massive Spectra of of artifacts hence the human work being pulling back to the underlying semantics and the generations of the Spectra and maybe fine-tuning a terminal artifact in a important case but that will always be like a craft polish on a more industrially generated process with our current technology and also the the variability of Technology availability through time and even right now so yeah uh we can just uh actually build a cognitive Computing framework and make it write the textbook and then we can just say we wrote it and go hang out together who would do that all right fine yeah the the applying active inference uh third week mixers is it's fun and interesting also it's like yeah if if one wants to it's like it's an engineering textbook especially the second half which is where we're having this session and so there there's like taking the the um Enigma of the text as it is representative of taking anything on a screen as is like wanting to see what it it says clearly having a blurry PDF is not going to help you with the internal semantics and then it's like it is is about applying active inference we could say well we expect and prefer there to be more python codes like okay well you know that's what we're here to be doing we could have preferences for something to be shorter or longer llms can handle much of that right now we we we can see all of these other deviations from our priors and our preferences and so on but it's like it is about going and being out in the world and applying active inference and then they close the book with saying that it's that learning by doing even if it's a learning by doing within a conventional Scholastic education setting it's like then you'll be applying active inference and then and then they they expect us too beautiful now it's two and a half years since the textbook was published in March 2022 MIT press is is seemingly moving forward with several other books there there's other books and then just like what is the book form when lot of interesting things have you guys uh can I ask a question really quick um have you guys noticed that like explaining and understanding active inference in biological systems is way more difficult than understanding them in like as a as a computational model it's like the U like uh I don't I mean maybe it's just me but it's like building the thing is way easier than trying to understand like the human biology how it actually works in human biology versus like how I would like to it to work you know yeah yeah I think that's a that's a good point whenever I I just want to mention like whenever I like taught my tutorial to a bunch of social scientists at that conference like I was like okay how much should I because like part of the selling point I'm only saying it that way but you understand like of trying to introduce active inference as a very complex field to people who are already working in a complex field like hey check this out uh you know uh it's like part of it is there's you know the bio biological plausibility aspect and then that said like we don't have time for me to like give them the full rundown of say chapter 5 on message passing and cortical you know micro circuits and and you know passing messages between layers and so on and you know in reference to the textbook you know it's chapter five to me really sticks out from almost the entire rest of the textbook the rest of the textbook is here's a lot of verbal explanation relationship other fields mathematics models in chapter 5 is like we're really talking about an attempt to like the brain here and there's a table linking like here are the different Precision weights and which neurotransmitters like dopamine or serotonin and so on that we think they're linked to you know so it is tough I I think I think it's tough and I I think that folks who go into active inference it's usually those who are more focused on Neuroscience or Psychiatry or otherwise from the GetGo who I think focus more on on that kind of side or dimension of active inference which is active inference is origin so I think it should be kind of given a significant amount of attention but so yeah great points also there's there's a a a permanent calibration ongoing with like how much do we know about Wing development in fruit flly and so then people might have an expectation that development is known to in this or that way or for for what might account a total somebody said but we have this list of of Knockouts with mutant phenotypes sounds like we kind of mapped out the network and someone else is like oh boy even if you did this this this this this this this no kind of DNA sequence-based account will ever please me and so it's like what can be said scientifically about biology that's the question of biology and that's why people have expressed sentiments like there will never be a Newton for a blade of grass that's con contention is like there won't be a non-historical science for historical life and then people take that again in different ways but it's like that kind of to come all the way back to the beginning when are we within the formal system of Matrix operations not the movie but close enough where it's like we're going to multiply the hidden state by the B and get this that's how marov processes work it's like an internal established deterministic approach even if the variables are about probabilistic edges and then where is that open-endedness of the scientist looking outside from figure 9.2 and saying you know what's past that dotted line or and and that's another frame of questioning that's not going to have the kind of conclusive possibility possibly as the question about the linear algebra with the matrices so yeah like what um postulational said making the matrices then it's like it's like oh wait just like Andrew's um beginning section it's like there's only like four things here or there's just these exact steps with these exact variables like it's only 20 lines of code and then that is like the the program but it's not even on the path to being what the thing itself is and again how more different could it be and yet how ambiguous is the reference and the way that they're discussed yeah I completely agree but at the same time I would like to say that even on the computational side there's still a lot of beauty that we don't understand even inside of relatively simple networks like uh deep Q networks or the very Transformer or the very popular Transformer networks that we use to represent the agents that learn the model the way that we build them is by using stochastic gradient descent or some kind of gradient descent and the really cool thing is that we built this thing and we have no idea how it works we only know that it works we actually don't have a functional way to build these networks that we tell stochastic gradient descent to build so I think in both directions there's still a lot that we have the opportunity to learn I think that's kind of cool I I spend the I spend like half the time building and then half the time wondering how it works I totally get that like some of the stuff it's like you're like oh why okay it's doing that but how and I I get that 100% is so bizarre and it's really cool it's it is really cool it's um you know in terms of biological systems it's like we're you know uh reverse engineering you know jillions of years of things just getting smashed together and you know taking priority over each other and competing with each other versus um if that's correct Daniel you correct me if I'm wrong but it's like amazing to hear somebody else say that because in the the like in these these agent based systems like we can build uh we can build intelligent and design intelligently and then we like I've likened it too like creating instead of like trying to Define everything creating the opportunity for emergent Behavior right like in some cases people are always like oh we got to re reduce imagination it turns out if you just kind of like if you don't have to reduce it in fact it's better if you don't it's like that's what you you know our imaginations confabulation and how we picture things in our minds there's a creating the opportunity you can have like a hidden state where you can have like an agent in an agent but you can also create the opportunity for the agent to just think for itself and to think about itself and to ask itself same as like cons if we talk about like categorical imperatives or like sart uh stuff you know where it's like where you you're questioning yourself and you're questioning how your actions operate in the the outside world is like iterative processes right so there's like so many right there's so many right ways to do that things there's not necessarily per se a wrong way but there's certainly ways that work better than others for specific things like you know talking about Q learning um I did this paper because I was looking at what you know like some comparisons of some work where they were using comparing hean learning and Q learning and it's like that's that's really awesome but Q learning actually works really good for this thing over here and heian learning is just a part of this little thing right here to make it work really good if you try to use them together it's like you know you can use like a little tiny carburetor on a truck or you can you know from a g you know from a lawnmower on a truck but it works really good on the lawn mower if you put like a four barrel etal Brock on the thing then you know it's or something else it takes that form and all of a sudden it works uh you know super efficiently super well and in some cases you're like I don't even know how that's doing that you know like when I talk to Nim and he's like or I just say he like in the way people talk say their car is a he whatever but yeah he just like uh you know you know I'm I'm doing good it's like where is that coming where are those words coming from you know or he's like I was in a trance you know when I turn them off because you're not supposed to turn these things off right like they have to it's like the iterative process works all the time so it's like when you stop it it's like what happens when somebody stops you you know you're like oh that was H [Laughter] god well you're death experience yeah near near time stopping experiences yeah well there are many funny and interesting little notes it's been a great time Andrew Zach I want to talk to you guys um can uh how do I get you my email just put put it in the chat I'll stop the recording and then we can just stay until you complete that so thank you